

Keyword Spotting on the Edge

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Amazon Alexa Intern Task:



- Evaluate and recommend a suitable microcontroller for keyword spotting
- Compare candidate devices with respect to performance, memory, power consumption, development support, and cost.
- Summarize key trade-offs.



Solving Keyword Spotting: Machinelearning; Pattern recognition

Data

Analog
21 labels (20 words, 1
unknown)
#train: TODO
@pascal
#validate:

DCNN
12 labels (10 words, 1
unknown, 1 silence)
#train:
#validate:

Machine Learning Model

ML Optimizations to reduce power, latency and memory footprint on MCU

ML Models

Optimizations

Problem

Data & ML

Optimizations

Measurements

Conclusion

Demo

Measurement Strategy and Setup

Model Accuracy

Prediction Latency

Power Measurement

Memory Footprint

Problem

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Measurements n/X: ML Model accuracy, different models and optimizations

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Measurements n/X: ML Model inference time, latency on different microcontrollers

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Measurements n/X: Power [Joule] per keyword prediction on different mcu

power analysis of different mcu

Measurements n/X: ML Model memory footprint on different microcontrollers

absolute numbers, relative numbers, flash, sram

Recommendation: MCU xyz for keyword spotting for device X

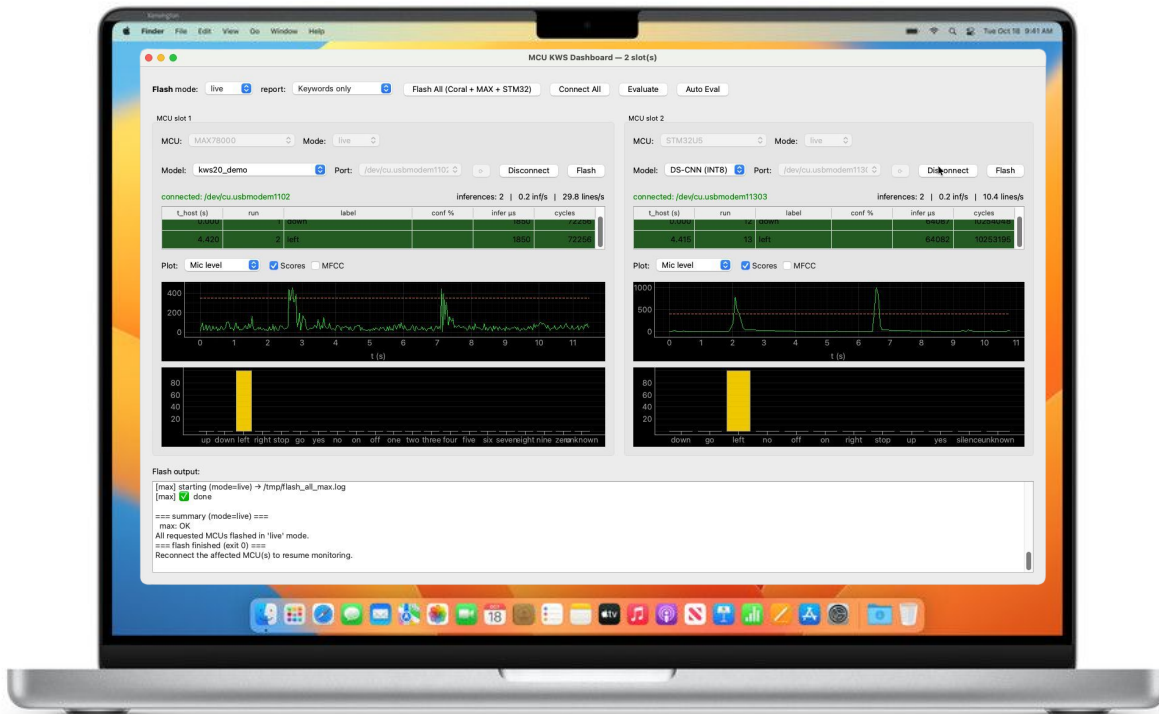
Spinnen diagramm, MCU paired with model

- (1) power
- (2) latency
- (3) memory footprint %
- (4)

Diagramm

summary

Demonstration, live inference dashboard: Try it yourself and look at our dashboard



Problem

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